

A Soft Introduction to CMA-ES Technique

Optimization Algorithm Team

Seangleng Khe (KHM)

Aphinan Photun (Thai)

Phoebe Jean Domen (Filipino)

Luong Minh Tuan (Viet)

Bui Huynh Xuan Quyen (Viet)

Instructor

Prof. Gaël Raoul

September 10, 2026

SEAMS School 2026: Applying Mathematics to Real-World Applications
HO Chi Minh City University of Science, Vietnam

Introduction

Optimization with Gradient Descents

Random Search

Evolution Strategy: (μ, λ) -ES

Ill-Conditioned Quadratic

Introduction

The objective of the study is to understand the progression

gradient descent → random search → evolution strategy → CMA-ES

After: we shall see how important it is to adapt a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

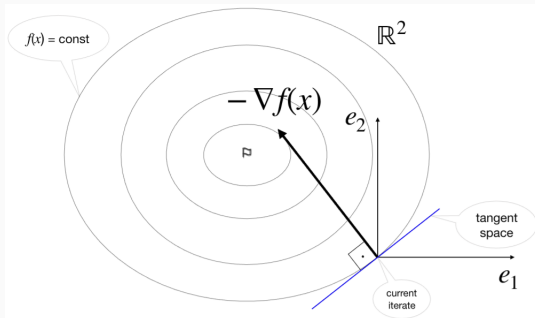
Optimization with Gradient Descents

Suppose we want to solve

$$\min_{x \in D} f(x)$$

If f is **smooth**, we use
gradient information

$$x_{n+1} = x_n - \sigma \nabla f(x_n)$$



Search information

The optimization algorithm uses gradient to search for the optimizer.

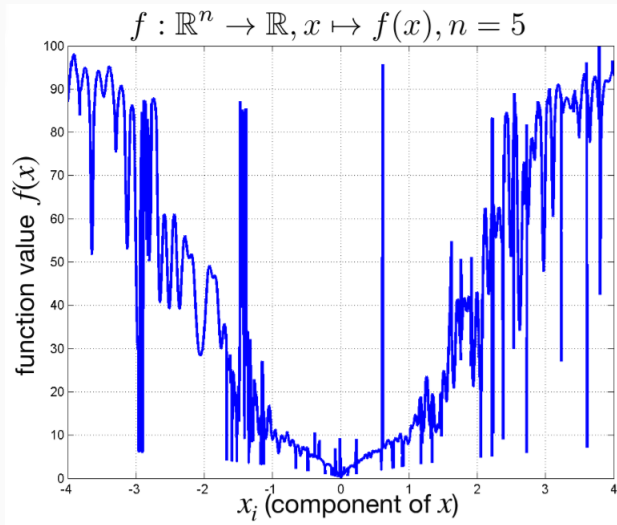


Figure 1: 5D Rugged Landscape ¹

¹Hansen [2019]

Random Search

How can we reach the optimizer?

Random Search ²

With a starting point $m_0 \Rightarrow$ generate random points around $m_0 \Rightarrow$ calculate the function values f

Search information

The optimization algorithm uses a set of randomly generated points based on probability distribution and calculate the fitness value (function value).

This is as stated " Finding a needle in an n-dimensional haystack " ³.

²White [1970]

³Korn [1966]

Evolution Strategy: (μ, λ) -ES

From random search, we can

Algorithm 1: $(1, \lambda)$ -ES Iteration: Isotropic Search

Input: A mean $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ

```
1 for  $k \leftarrow 1$  to  $n$  do
2   for  $i \leftarrow 1$  to  $\lambda$  do
3     Sample  $Z_i \sim \mathcal{N}(0, I_d)$ 
4     Compute  $X_i \leftarrow m_k + \sigma Z_i$            // offsprings  $X_i \sim \mathcal{N}(m_0, \sigma^2 I_d)$ 
5   Compute function value  $f(X_1), \dots, f(X_\lambda)$ 
6    $i^* \leftarrow \arg \min_{i \in \{1, \dots, \lambda\}} f(X_i)$  // Choose one parent
7
8    $m_{k+1} \leftarrow X_{i^*}$ 
9 return new mean  $m_n$ 
```

Algorithm 2: (μ, λ) -ES Iteration

Input: A point $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ , number of selected parent $\mu \leq \lambda$, weighted factor ω_i s.t. $\sum_{i=1}^{\mu} \omega_i = 1$.

1 **for** $k \leftarrow 1$ **to** n **do**

2 **for** $i \leftarrow 1$ **to** λ **do**

3 $Z_i \sim \mathcal{N}(0, I_d)$

4 $X_i = m_k + \sigma Z_i$

5 Compute $f(X_i)$ and sort offspring by fitness:

$$X_{1:\lambda}, X_{2:\lambda}, \dots, X_{\lambda:\lambda},$$

where $X_{1:\lambda}$ is the best, $X_{2:\lambda}$ is the second-best, and so on.

6 Choose μ best candidates and recombine

$$m_{n+1} = \sum_{i=1}^{\mu} \omega_i X_{i:\lambda}.$$

Updated Search Information

The optimization algorithm a starting point m to λ randomly generated points (called *offsprings*) and selects μ points (called *parents*), recombine to generate a new points, and repeat the process.

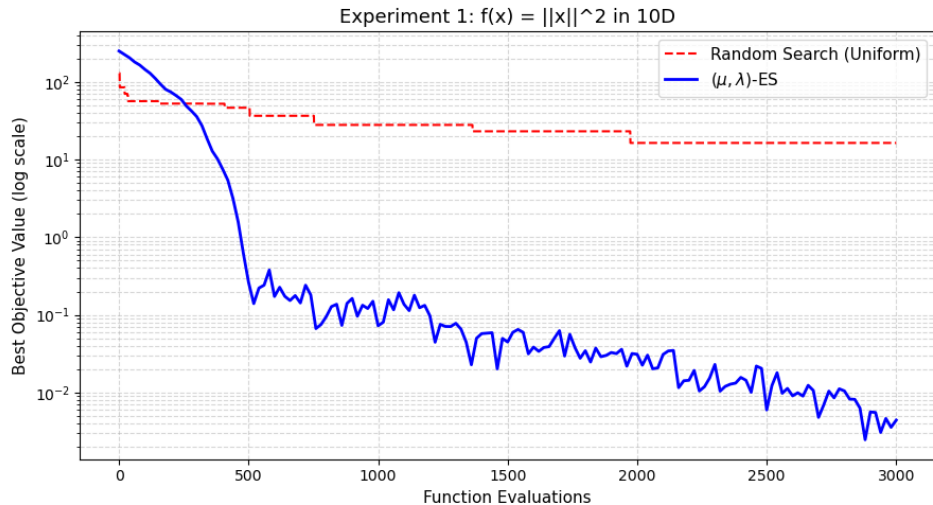


Figure 2: Comparison between random search with uniform distribution and (μ, λ) -ES.

III-Conditioned Quadratic

(μ, λ) -ES is not enough...

- It more or less is taking a detour to obtain the solution.
- A central observation is that isotropic mutation is inefficient when the successful search from a thin elongated ellipsoid.

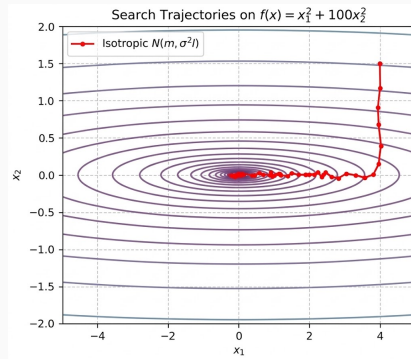
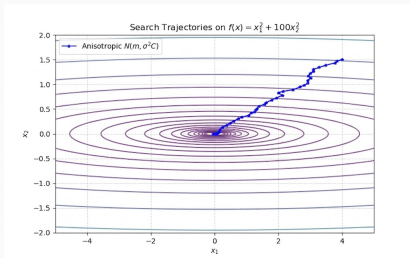
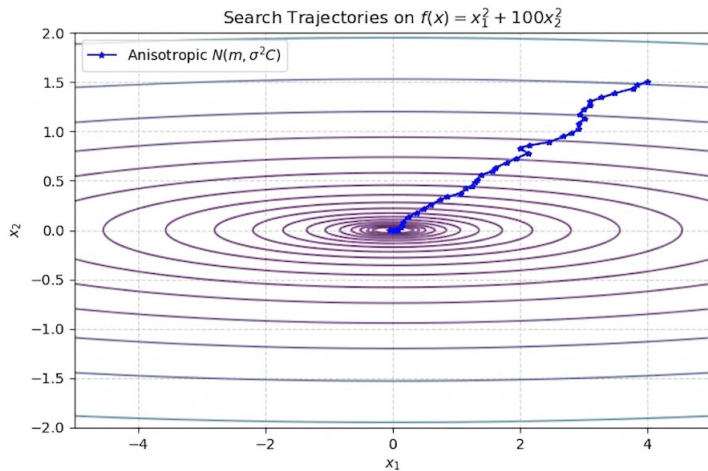


Figure 3: Function f has anisotropic level set. (Steep valey in x_2 and flat in x_1).



Anisotropic search

In stead of sampling from

$$X \sim \mathcal{N}(m, \sigma^2 I),$$

we consider the offspring of the form

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

What does the covariance matrix C do here?

A picture here similar to the one that prof explained in the first day.

The picture equals =

- The matrix C determines the shape of and orientation of the search distribution.
- The idea of covariance is relatively simple: Directions in which successful steps have repeatedly occurred should receive larger variance in the future generations.

Search Information

The optimization algorithm learns not only where to search though the starting point m but also how to search through C .

References

- Nikolaus Hansen. How to evolve gradient descent into evolution strategies and CMA-ES. Presentation slides, Symposium on Evolutionary Algorithms: Back to the Future and Beyond, Delft, Netherlands, September 2019. URL <http://www.cmap.polytechnique.fr/~nikolaus.hansen/cmaes-from-gradient-descent-delft2019.pdf>.
- R.C. White. *A survey of random methods for parameter optimization*. EUT report. E, Fac. of Electrical Engineering. Technische Hogeschool Eindhoven, 1970.
- Granino Arthur Korn. *Random-Process Simulation and Measurements*. McGraw-Hill, New York, 1966.

**Thank you for your attentions!!!
:-)**